

1. **Experiment:** Macrophage Polarization Detection of STAR-Trap
2. **Time:** 2026.07.31-2026.08.12
3. **Member:** Yue Fu, Haochen Xu, Xinru Yuan, Kaiqing Zhang, Qichang Liu, Yang Jin
4. **Materials:** STAR-Trap macrophages, WT-M ϕ control macrophages, human Fc receptor-blocking reagent (BD Biosciences), fluorophore-conjugated antibodies: anti-CD86 (clone BU63, BD Biosciences, cat. no. 572835), anti-HLA-DR (clone L243, BD Biosciences, cat. no. 568231), anti-CD206 (clone 15-2 [MR15-2], BD Biosciences, cat. no. 572493), anti-Arg-1 (clone 7D1D10, BD Biosciences, cat. no. 572285), anti-IL-10 (BD Biosciences, cat. no. 567013), BD Perm/Wash buffer system (cat. no. 554723), CM-H2DCFDA probe (Beyotime, cat. no. S0035M), FACS buffer (PBS + 1% FBS), isotype controls, flow cytometer, FlowJo software.

5. Method:

- (1) Cell collection and Fc receptor blocking:

STAR-Trap and WT-M ϕ cells were collected and washed once with ice-cold PBS. Cells were resuspended in FACS buffer and incubated with human Fc receptor-blocking reagent (BD Biosciences) for 15 min at 4 °C.

- (2) Cell-surface staining:

For surface markers, cells were incubated for 30 min at 4 °C in the dark with fluorophore-conjugated antibodies against CD86, HLA-DR, and CD206. After incubation, cells were washed twice with FACS buffer and resuspended for further staining or acquisition.

- (3) Intracellular staining of Arg-1 and IL-10:

Cells were fixed and permeabilized using the BD Perm/Wash buffer system according to the manufacturer's instructions. Then cells were incubated with anti-Arg-1 and anti-IL-10 antibodies for 30 min at 4 °C in the dark. After washing, cells were resuspended in FACS buffer.

- (4) Intracellular ROS detection:

Intracellular ROS were detected using the CM-H2DCFDA probe (Beyotime) following the manufacturer's protocol. Briefly, cells were incubated with the probe at 37 °C for 30 min in the dark, washed, and resuspended in FACS buffer.

- (5) Flow cytometry acquisition and analysis:

Stained cells were analyzed by flow cytometry. At least 50,000 events were recorded per sample. Data were processed using FlowJo software. Isotype controls were used to establish gating boundaries. The gating strategy included: FSC-A/SSC-A for cell population, FSC-A/FSC-H for singletons, viability dye for live cells, and then analysis of HLA-DR, CD86, ROS, CD206, Arg-1, and IL-10 expression. The percentage of positive cells and mean fluorescence intensity (MFI) were recorded for each group.

- (6) Statistical analysis:

Flow cytometry data were analyzed using FlowJo software. Median fluorescence intensity (Median FI) values are shown for one representative experiment.

6. Result:

Compared with WT-M ϕ , STAR-Trap macrophages showed increased expression of M1 markers HLA-DR (MFI: 1428 vs 984) and CD86 (MFI: 2918 vs 1765), as well as elevated ROS levels (MFI: 3758 vs 1670). In contrast, M2 markers CD206 (MFI: 1031 vs 3785), Arg-1 (MFI: 3090 vs 3192), and IL-10 (MFI: 2948 vs 3520) were downregulated in STAR-Trap cells. These results indicate that STAR-Trap macrophages are polarized toward an M1-like phenotype.

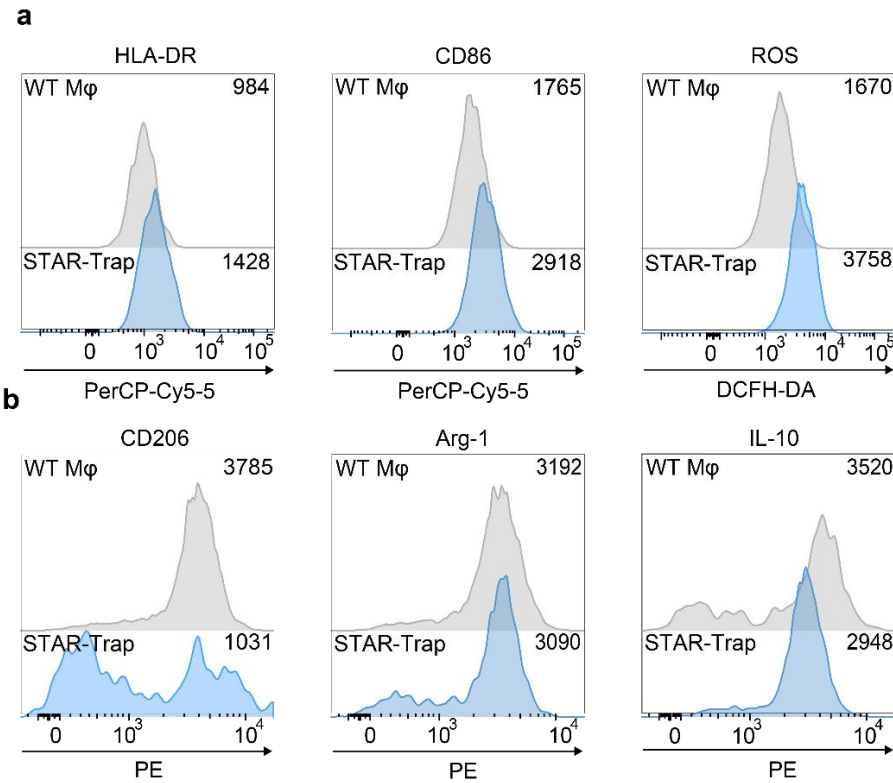


Figure 1. STAR-Trap exhibit an M1-like phenotypic shift. a, Representative flow-cytometry histograms showing the expression of the M1-associated markers HLA-DR and CD86 and intracellular ROS levels in wild-type macrophages (WT Mφ) and STAR-Trap. b, Representative flow-cytometry histograms showing the expression of the M2-associated markers CD206, Arg-1 and IL-10 in WT and STAR-Trap. Grey histograms represent WT macrophages, and blue histograms represent STAR-Trap. Numbers indicate the median fluorescence intensity of the corresponding cell populations.